What makes a matching market congested?

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Stanford & Oxford

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- We call this **congestion** and explore its causes and remedies

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Focus on the simplest model that gives rise to congestion.

Setting: decentralized matching

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One-to-one matching: at most one match per applicant and per firm

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- Then vary preference alignment and time constraints

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- Low application costs increase the chance a firm gets some applications
  - E.g. online dating platforms, economics job market, LinkedIn
- But setting application limits decreases the chance that offers go to the same agents
  - E.g. limit the number of swipes, send special signals, limited messaging

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- Focus on this tradeoff via **quotas**
  - Additional results: time constraints and preference correlation

Application and offer graphs

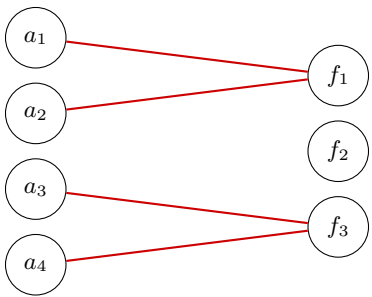
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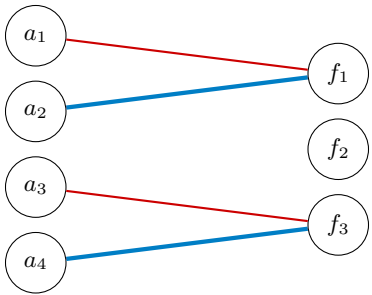


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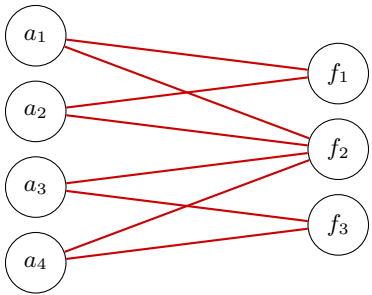


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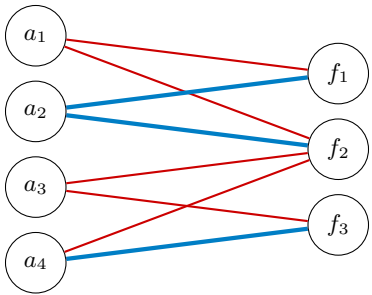


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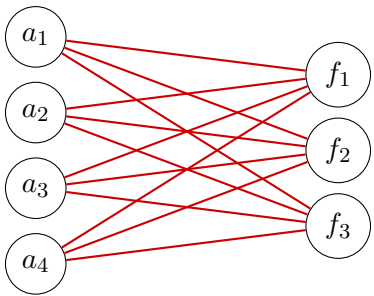


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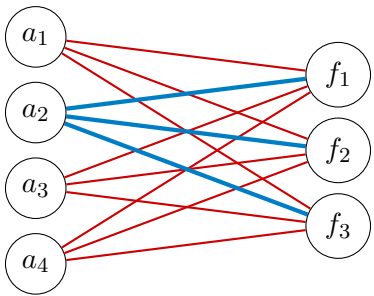


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- Harder to prove in finite markets: offers become correlated even under i.i.d. values

Results Preview (2)

Baseline market

i.i.d. values $A \geq F$

One-to-one matching One round

Robust

Not robust

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- Preference correlation in matching:
  - Castera et al 25 shows it can improve matching
  - Contribution: worsens congestion without market clearing

Model & Results

Model: Independent values

$$A \geq F$$

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$A \geq F$ and one-to-one matching

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On each side: continuous CDF, strictly positive support

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Equilibrium

- Unique symmetric equilibrium:
  - Apply to q best firms
  - Offer best applicant
  - Accept best offer

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In all the markets we will study, the following is true:

- If $q < q'$ and q has a higher match rate then it has higher applicant utility

Independent values

Proposition

If $A \geq F$ then $q = 1$ has a higher match rate than $q = F$

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- Why is $q = F$ tractable? Equivalent to firms applying with $q = 1$

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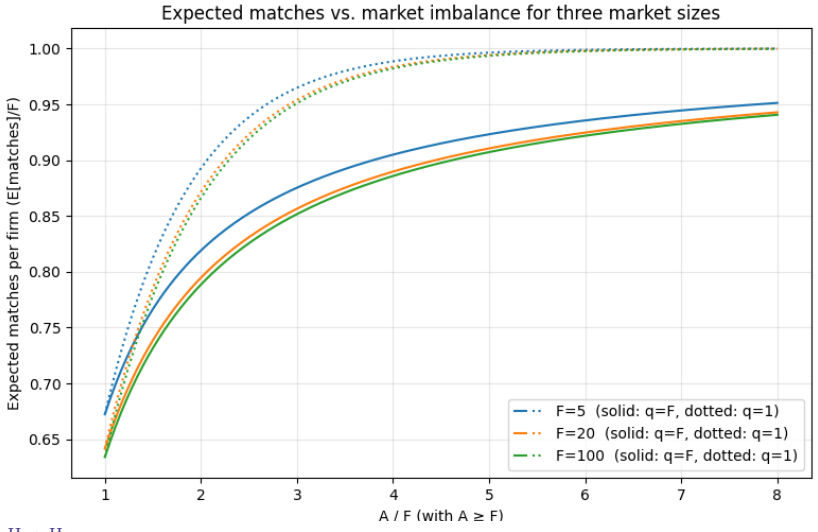
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Independent values: 1-Quota vs Full Application

▸ Proof



Independent values: optimal quota

Theorem

In the i.i.d. environment,

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2. *if* $A \geq F$, *then* $q^* \leq 3$.

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- Large markets, offers are independent $\Rightarrow$ easier to see

Independent values: large-market intuition

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If our applicant applies to this firm, her probability of receiving its offer is

$$p_q = \mathbb{E}\left[\frac{1}{K+1}\right] = \frac{1 - e^{-\lambda q}}{\lambda q}.$$

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- $\log(1 - p_q) < 0$: another application gives another chance to match.
- $p_q' < 0$: when everyone sends more applications, each application becomes less likely to succeed.

Independent values: how congested should the market be?

Recall

$$p_q = \frac{1 - e^{-\lambda q}}{\lambda q}.$$

Both the coverage gain and collision cost depend on q and λ only through λq, expected application received *per firm*

The two forces balance at approximately

$$\boxed{\lambda q^* \simeq 2.66}.$$

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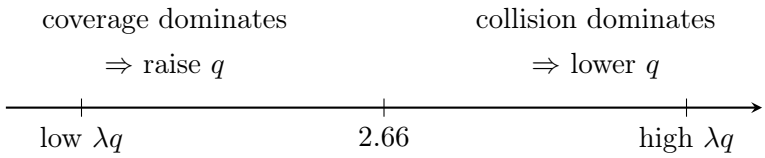
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- The improvement varies by quota, so we cannot simply use the large-market benchmark as a bound

Independent values: finite-market dependence

Consider two firms to which the same applicant has applied

- Under independence, the probability of receiving both offers is p_q^2

Here $p_q = \dfrac{F}{Aq}\left[1 - (1 - q/F)^A\right]$ is the finite-market offer probability

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- Competitors apply to distinct firms: applying to one makes them less likely to apply to the other
- Hence

$$\Pr(\text{receiving both offers}) \leq p_q^2$$

Finite sampling mitigates collision

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Finite market: proof roadmap

The bounds are $\boldsymbol{L_q := 1 - (1 - p_q)^q}$,

When $A \geq 2F$

When $F \leq A < 2F$

Finite market: proof roadmap

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- Bound finite joint offers against independence by δ_q

$$b_q := \Pr(O_1 \cap O_2)$$

$$0 \leq \underbrace{p_q^2}_{\substack{\text{two offers if} \\ \text{independent}}} - \underbrace{b_q}_{\substack{\text{actual two-offer} \\ \text{probability}}} \leq \delta_q = \frac{F^2}{q^3(A-1)A^2}.$$

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Fewer shared rivals make joint offers less likely.

Finite market: where δ_q comes from

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- Sum over $A-1$ rivals, then relax the factors

$$0 \le p_q^2 - b_q \le \underbrace{(A-1)}_{\text{competitors}} \cdot \underbrace{\Delta_{2,q}}_{\text{overlap gap}} \cdot \underbrace{\kappa_q^2}_{\text{two-firm pivotality}} \le \underbrace{\frac{F^2}{q^3(A-1)A^2}}_{\delta_q}.$$

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This bound is loose: $q = F$ gives $p_q^2 - b_q = 0 < \delta_q$.

Finite market: summary and caveats

- Small benchmarks make pairwise offer bounds enough

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- Larger benchmarks need higher-order joint offers, hard to bound
- $q^* \le 3$: analytic for $F \ge 225$; exact rational brute force for $4 \le F \le 224$

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- When $A \geq F$, *very small* quotas maximize matching
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- Next: three extensions, then preference correlation
  1. $F \geq A$
  2. Many-to-one matching
  3. A second round

When firms outnumber applicants

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Proposition (Large market)

If $A \leq F$, then

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- Ex: 300 firms and 100 applicants, no more than 10 app per person

Even when applicants are scarce, optimal quotas remain bounded.

Many-to-one

What about a setting with more “seats” per schools?

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In the many-to-one large matching market, if $c \geq 2$ *then* $q^* = 1$.

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Proposition

In the many-to-one large matching market, if $c \geq 2$ then $q^* = 1$.

- Increasing capacity gives applicants a greater chance of admission!

Second Round Value

- Recap:
  - Quotas are very impactful: efficiency gain of up to 10%
  - Optimal quotas is usually just 1
- Quotas can be seen as, arguably, restrictive - what about a second round?

Second Round Game

- We consider the game with full application but with an additional clearing market:
  1. Applicants apply to ***all*** firms
  2. Firms receive applications and each makes 1 offer
  3. Applicants receive offers and ***hold*** one - rest rejected
  4. ***Rejected firms make an additional offer***
  5. Applicants see all offers, ***held and new***, accept one

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Theorem

If $A \geq F$ *then*

$$MR(2 \text{ rounds}) \geq MR(\text{quota of 2 on firms})$$

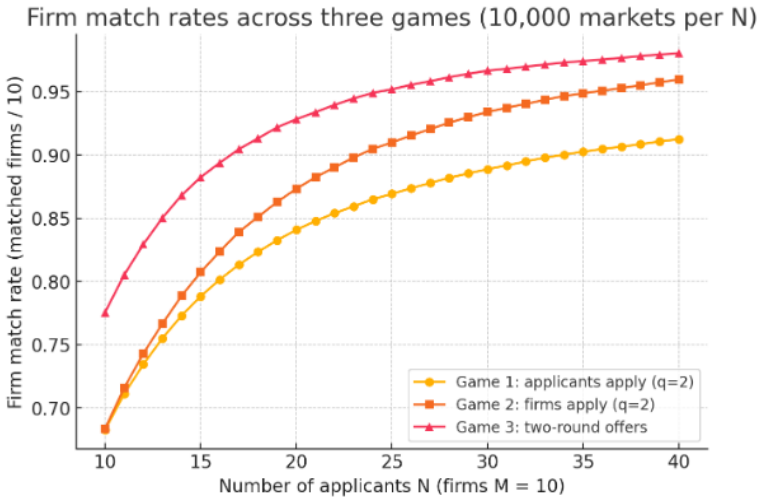
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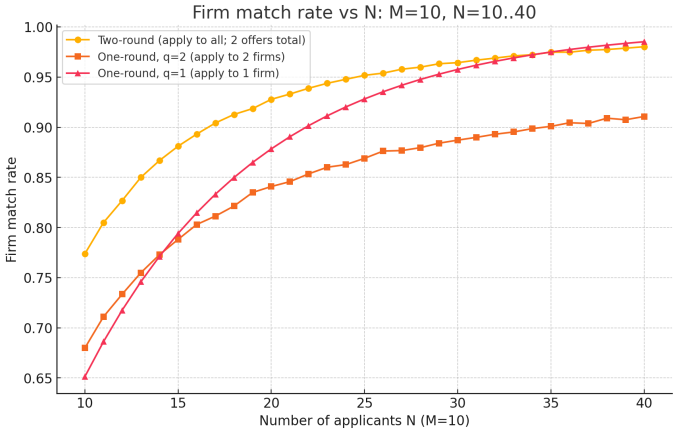
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Firm match rates across three games (10,000 markets per N)
Firm match rate (matched firms / 10)
0.95
0.90
0.85
0.80
0.75
0.70
10
15
20
25
30
35
40
Number of applicants N (firms M = 10)
Game 1: applicants apply (q=2)
Game 2: firms apply (q=2)
Game 3: two-round offers



- Most of the gains are captured by just one additional round

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- On the flip side: power is commensurate with cost of additional time

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- We show it can be more subtle

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$\mathbb{E}[\#$ of matches$]$	$=$	$\mathbb{E}[\#$ of firms w/ application$]$	$-$	$\mathbb{E}[\#$ of wasted offers$]$
"match rate"		"coverage"		"collision"
$\nearrow$ quota q		$\nearrow$		$\nearrow$
$\nearrow \beta^a$		$\searrow$		depends
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Model: Correlation

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Cross-side alignment: what changes?

- A shared component aligns applicant and firm values

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- Harder: increase in collisions

Cross-side alignment: endogenous stars

- Assume $u_{af} = u_{fa}$ - fully shared values

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   - Cross-side alignment: also learn your value was not the highest
     - ★ Values are iid, but top q values are positively associated
     - ★ An opponent had a higher value than you
2. What does this mean for your other application:
   - That application likely has low value (persistent in large market)
   - If you run into the same opponent, it's bad news (fade away in large market)

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- As signaling gets better, more congestion on top candidates
- Highlighting a congestion (quantity) vs match value (quality) tradeoff

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  - Under a quota of 2 : $m_2^* = 0.24 > 0.23 = m_2^{iid}$, why?
- Applicant correlation can concentrate some applications and thus make the rest of the market behave as if under a lower quota.

Main result: Correlation

Let $m_q^{i.i.d.}, m_q^{\alpha}, m_q^{\beta^f}$ and $m_q^{\beta^a}$ be the expected match rates in the aligned environments for a fixed q

Theorem

The following are true for every fixed q:

1. $m_q^{i.i.d.} \geq m_q^{\alpha}$.
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Theorem

The following are true for every fixed q:

1. $m_q^{i.i.d.} \geq m_q^{\alpha}$.
2. $m_q^{i.i.d.} \geq m_q^{\beta^f}$.
3. *Let $N_1, ..., N_q$ denote the number of competitors at the various firms that an arbitrary applicant faces in the applicant-aligned environment.* $m_q^{i.i.d.} \geq m_q^{\beta^a}$ *if*

$$\mathbb{E}\left[\prod_{f=1}^{q} \frac{N_f}{N_f+1}\right] \geq \prod_{f=1}^{q} \mathbb{E}\left[\frac{N_f}{N_f+1}\right].$$

Conclusion

The ***coverage-collision*** tradeoff

- More applications ⇒ more coverage ⇒ but more collisions
- Small quotas are *typically* optimal
- We don't see this as prescriptive — illustrative of what *congestion* is

Additional results on: many-to-one, time constraints, correlation

- None of which surprising, expost - align with intuition about congestion

Future Agenda

Starting point for a rich research agenda

- Abstracted away from strategic concerns
- Quota vs Application cost as design tools
- Turnaround time vs Total time
- Experiments + Empirics

Thank you :)